**INCH POUND**

MIL-STD-1504D
16 May 2022

SUPERCEDING
MIL-STD-1504C
01 March 2007

# DEPARTMENT OF DEFENSE
# STANDARD PRACTICE

## ABRASIVE BLASTING

Reactivated after 16 May 2022 and may be used for new and existing designs and acquisitions.

AMSC: NA

AREA: MFFP

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# FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. This standard provides guidance for the repair process, acquisition, and manufacture of parts and/or spare parts on the landing gear of all military aircraft.

3. Comments, suggestions, or questions on this standard should be addressed to: DAF 309th CMXG 5760 Southgate Ave. Bldg. 511 Hill AFB UT 84056-5228, email 309.CMXG.Workflow@us.af.mil; DAF Landing Gear Program, Attn: Engineering at AFMC 417 SCMS/GUEA, Hill AFB, UT 84056-5228, email 417SCMS.GUEA.workflow@us.af.mil. Since contact information can change, verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.



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SUMMARY OF VERSION D MODIFICATIONS

1. Changed the Uniform Resource Locator (URL) for the Acquisition Streamlining and Standardization Information System (ASSIST) database throughout the document.

2. Updated foreword adding updated contact information.

3. Updated custodian and review activities in concluding material

4. The following modifications to MIL-STD-1504D have been made:

| PARAGRAPH | | MODIFICATIONS | |
|---|---|---|---|
| | Title page | | Updated |
| | Forward | | Updated |
| | Contents pages | | Updated |
| | Change summary page | | Added |
| | 1.1 | | Changed |
| | 2.2.1 | | Changed |
| | 2.2.2 | | Changed |
| | 4.1 – 4.4 | | Changed |
| | 4.1.1 – 4.4.2 | | Added |
| | 4.5 | | Changed |
| | 4.6.1 b., g., h., and NOTE | | Changed |
| | 4.6.2 a. – d. | | Changed/deleted |
| | 5.1 – 5.3 | | Changed |
| | 5.3.1 – 5.3.4.2 | | Added |
| | 5.4 – 5.8 | | Changed |
| | 5.8.1 – 5.8.4 | | Added |
| | 5.9 all | | Moved (mis-numbered in version C) |
| | 6.1 – 6.3 | | Changed |
| | CONCLUDING MATERIAL | | Changed |



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# 1. SCOPE

1.1 Scope. This standard specifies the processes and materials required for abrasive blasting of military aircraft parts and missile components. Abrasive blast cleaning consists of propelling a high-pressure stream of abrasive media against a surface. This document may be used for other parts or applications if required, and or approved by the Activity's cognizant engineering authority.

1.2 Abrasive blast use. Typical uses include, but are not limited to:

a. Corrosion removal.
b. Conditioning of surfaces for subsequent finishing.
c. Removal of coatings, scale, dry soils, or other contaminants.
d. Smooth or roughen a surface.

# 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3. 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and commercial item descriptions. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

| | |
|---|---|
| MIL-DTL-5002 | Surface Treatments and Inorganic Coatings For Metal Surfaces of Weapons Systems |
| MIL-PRF-9954 | Glass Beads: for Cleaning and Peening |
| MIL-DTL-85891 | Engineered Media for Removal of Organic Coatings |

FEDERAL SPECIFICATIONS

| | |
|---|---|
| TT-C-490 | Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings) |

COMMERCIAL ITEM DESCRIPTIONS (CID)

| | |
|---|---|
| A-A-1722 | Grain, Abrasive (Soft Blasting) |
| A-A-59316 | Abrasive Materials; for Blasting |

(Copies of these documents are available online at https://assist.dla.mil)



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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

| | |
|---|---|
| 29 CFR §1910 subpart Z | Occupational Safety and Health Standards (OSHA)<br>Toxic and Hazardous Substances |

(A copy of this document is available online at https://www.osha.gov/.)

2.3 Non-Government publications. Not applicable.

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this standard and the references cited herein, the text of this standard takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Not applicable.

4. GENERAL REQUIREMENTS

4.1 Abrasive blasting. Abrasive blasting shall not be used on aircraft components unless specifically authorized by the applicable drawings, process specifications, or approval by the Activity's cognizant engineering authority.

4.2 Cleaning. Parts to be abrasive blasted shall be free of grease and oil. Cleaning shall be in accordance with MIL-DTL-5002 or TT-C-490. If hydrogen embrittling cleaning methods are used and steels with ultimate tensile strength greater than 180,000 psi are processed, the components must be baked for a minimum of four hours at 375 +/- 25 °F. Baking to begin within four hours of the embrittling cleaning step.

4.3 Surface defects. Significant surface defects shall not be removed by abrasive blasting. These defects shall be removed as specified by the Activity's cognizant engineering authority. For example, light surface corrosion may be removed by blasting, however corrosion pits may not.

4.4 Blasting method. Automatic equipment is recommended where possible. The discharge flow shall be smooth and continuous to prevent uneven erosion of surfaces. Excessive blasting of surfaces shall be avoided to prevent warpage and excessive removal of material.

4.4.1 Stream of blast media. The blast media stream shall be in-motion at all times. Dwelling in one location shall be avoided to prevent significant damage to the surface. Blasting time shall be limited to only as long as necessary to adequately clean the surface.


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4.4.2 Blasting one surface of a part. Blasting one (1) surface of a part by reflecting off another surface of the part is not acceptable as this method removes significant amounts of material from the reflecting surface. For example, if blasting an interior part's surface, the operator shall not blast (1) one surface to reflect the blast media toward another surface that is not accessible to the direct blast of media spray. When applicable, use a blast nozzle tip that fans out the media spray or otherwise reflects the media. For some curved interior surfaces it may more effective to use specialized rotary abrasive tools that can follow the contours of the part.

4.5 Corrosion. Blast cleaning exposes bare metal to the environment. Time between blasting and further processing shall be kept to a minimum to ensure compliance with subsequent operations. For example, many coating specifications limit the time between blasting and coating to four (4) hours. After blasting, handling and contact with the blasted surfaces shall be minimized. If blasted parts are handled, operator shall use clean gloves and ensure the part does not come into contact with contaminated surfaces.

4.6 Materials and equipment.

4.6.1 Materials. Materials used for abrasive blasting are as follows:

a. Silicon oxide. Silicon oxide (sand) meets the requirements of A-A-59316, Type II.

b. Aluminum oxide. Aluminum oxide shall meet the requirements of A-A-59316, Type I, composition of 95% or greater aluminum oxide and less than 1% silica is acceptable.

c. Garnet. Abrasive media meets the requirements of A-A-59316, Type IV.

d. Abrasive grit. Abrasive grit meets the requirements of A-A-1722.

e. Abrasive material. Abrasive material meets the requirements of A-A-59316.

f. Glass beads. Glass beads meets the requirements of MIL-PRF-9954.

g. Engineered Media. Engineered (plastic) media meet the requirements of MIL-DTL-85891. Media covered by this specification is intended as an abrasive blasting material for paint removal.

h. Reclaimed materials. The use of blast equipment that recycles a portion of the blast media is recommended.

NOTE: Materials may be expanded beyond the Grades listed in A-A-59316.

4.6.2 Equipment. Equipment requirements for abrasive blasting shall be as follows:

a. Dust removal. The equipment shall remove the dust formed during blasting and shall insure that the dust does not cause atmospheric contamination.

b. Classification. The equipment shall include a screening or classification process to remove undersize or broken abrasive particles, corrosion particles, and paint and metal particles.

5. DETAIL REQUIREMENTS

5.1 Masking. Sections or areas of a part that are not to be abrasive blasted shall be masked off using tapes, plugs, stoppers, custom masking devices, or any other product that will not damage the part being blasted.


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5.2 Compressed air. The compressed air used for blasting, blowing off dust, or drying parts for dry abrasive blasting shall be free from moisture, oil and, solid particles. The air shall be filtered at the point of use or be supplied by facilities designed to deliver clean air. Clean and replace the filters or maintain the equipment as necessary to meet these requirements. Developing and performing the procedures to verify air cleanliness shall be the responsibility of the facility performing the blasting operations.

5.3 General blasting instructions.

5.3.1 Distortion and warpage. Blasted parts shall show no evidence of distortion or warpage. Use care in blast cleaning to avoid excessive local blasting. Use fine abrasives and low air pressure on thin or low strength sections. The blast nozzle must always be kept moving over the surface of the part to avoid dwelling in one location. Blasting in one location can quickly remove excessive material, overheat the surface, and damage the part being blasted.

5.3.2 Complex geometries. The standoff distances and blast angles listed below are to be followed whenever possible, however parts often have complex shapes including interior surfaces where it is not physically possible to meet the standoff and angles specified. It is acceptable to blast the surfaces at different standoff distances and blast angles as needed based on the geometry of the part.

5.3.3 Media selection. When the media to be used is not specified, the media type and size shall be selected by the processor. It is recommended to use the least aggressive media available that will adequately perform the task. For example, engineered (plastic) media is the least aggressive in this specification. If engineered (plastic) media adequately removes paint, then engineered (plastic) media should be used. For coatings such as dry film lubricant or phosphate coatings, engineered (plastic) media is often inadequate, in which case glass media is the least aggressive option that will adequately remove the coating. If glass is insufficient, then an abrasive grit media such as aluminum oxide or garnet should be used.

5.3.4 Hand polishing.

5.3.4.1 Hand polishing is an acceptable alternative to blasting operations. This may be desirable in cases where the part is thin walled and easily damaged, or may be helpful to clean small areas that were not adequately blasted.

5.3.4.2 When hand polishing is performed, only soft backed abrasives shall be used. Hand power tools may be used with the soft backed abrasives if the tool is ½ horsepower or less using pneumatic power with 100 pounds per square inch (psi) maximum. Abrasives such as files are not soft backed and are not acceptable.

5.4 Recommended process flow.

a. Clean in accordance with paragraph 4.2.

b. Mask as required, paragraph 5.1.

c. Clean by abrasive blasting.

d. Clean with compressed air as required.

e. Continue processing as soon as possible.



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5.5 Glass bead blasting.

a. The nozzle should be held 8 to 31 centimeters (cm) (3 to 12 inches) from the surface.

b. The recommended air pressures for glass blasting:

| | |
|---|---|
| Ferrous Alloys | $2.8 \times 10^5$ to $4.2 \times 10^5$ Pa (40 to 60 pound force psi) |
| Aluminum Alloys | $2.1 \times 10^5$ to $3.5 \times 10^5$ Pa (30 to 50 psi) |
| Magnesium Alloys | $6.9 \times 10^4$ to $2.8 \times 10^5$ Pa (10 to 40 psi) |
| Titanium Alloys | $2.8 \times 10^5$ to $4.2 \times 10^5$ Pa (40 to 60 psi) |

5.6 Silicon oxide, aluminum oxide, and garnet blasting.

**WARNING**

**Avoid excessive inhalation of abrasive dust. Provide ventilation as required in accordance with 29 CFR § 1910 subpart Z.**

a. The nozzle shall be held at 15 to 62cm (6 to 24 inches) from the surface.

b. The air pressure shall not exceed $5.5 \times 10^5$ Pa (80 psi).

5.7 Engineered (plastic) media blasting. Engineered (plastic) media blasting is typically used for organic coating removal from metal surfaces.

**WARNING**

**Engineered (plastic) media blasting shall not be used on fiberglass or other composite materials or on metals having a thickness of less than 0.064 inches without the authorization of the responsible engineering authority. Fiberglass is typically sanded to remove organic coatings without excessively damaging the fiberglass.**

a. Use media conforming to the requirements of MIL-DTL-85891. Media shall have a particle size of US screen 10 to 60 mesh.

b. Pressure should be between 2.8 x 105 to 4.2 x 105 Pa (40 to 60 psi) at the blast nozzle for media having a Barcol hardness of 34 to 42 and less than 4.2 x 105 Pa (40 psi) at the nozzle for media having a Barcol hardness of 54 to 72.



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5.8 Quality control.

5.8.1 Quality control. The Activity's responsible quality control department shall enforce the requirements of this standard. Inspection to meet the requirements shall be performed with such frequency as deemed necessary by the quality control department to assure compliance with the standard.

5.8.2 Requirements. After abrasive blasting, the following requirements shall be satisfied:

a. Blasted surfaces shall be free of scale, paint, and corrosion products.

b. Blasted surfaces shall be free of dust, silt, powder, and other contamination.

c. Parts shall be free of warpage, distortion, and excessive material removal.

5.8.3 Rejected parts. Parts rejected due to warpage, distortion or excessive material removal shall be referred to the Activity's cognizant engineering authority for further action. Parts rejected because of insufficient blast cleaning shall be re-cleaned in accordance with this document.

5.8.4 Drawing requirements. Drawing requirements for surface finishes shall be met after the blast cleaning operation. When authorized by drawing, other applicable document, or by the Activity's cognizant engineering authority, light machining or grinding methods may be used to produce the required surface finish after blasting.

6. NOTES

This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. This intended use of this standard is to provide the processes and procedures used for abrasive blasting on all military aircraft's landing gear components. But, it may also be used for other abrasive blast operations when approved by the Activity's cognizant engineering authority.

6.2 Subject term (key word) listing.

Abrasive material
Aluminum oxide
Corrosion
Glass beads
Engineered (plastic) media
Silicone oxide
Surface defects

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes in version D. A *Summary Change* page has been added.



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CONCLUDING MATERIAL

Custodians:
Air Force – 70
Navy – SH

Preparing Activity
Air Force – 70
(Project MFFP-2020-006)

Review Activities:
Army – MR
Navy – AS
Air Force – 20

NOTE: The activities listed above were interested in this standard as of the date of this standard. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.


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